Tutorial 4 - Analogue to Digital Conversion

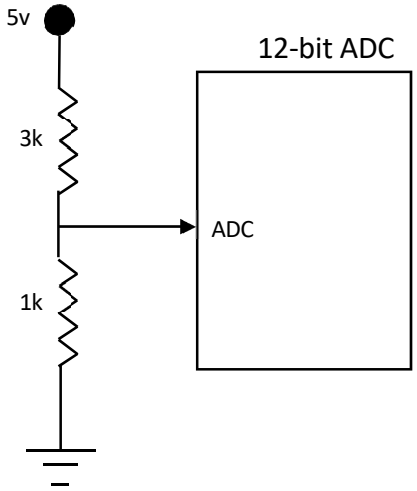
Q4.1 In a 12-bit ADC, the reference voltages are: VREF+ = 3.3V, VREF- = 0V. Estimate the digital result if the voltage at the analog input is 2.5V.

(a) 2047_{10}
(b) 775_{10}
(c) 3102_{10}
(d) 3412_{10}

2^12 - 1 = 4095
v = 2.5x(4095/3.3)
v = 3102.27(decimal)

C

Q4.2 Based on the circuit below, what is the voltage at the ADC pin?

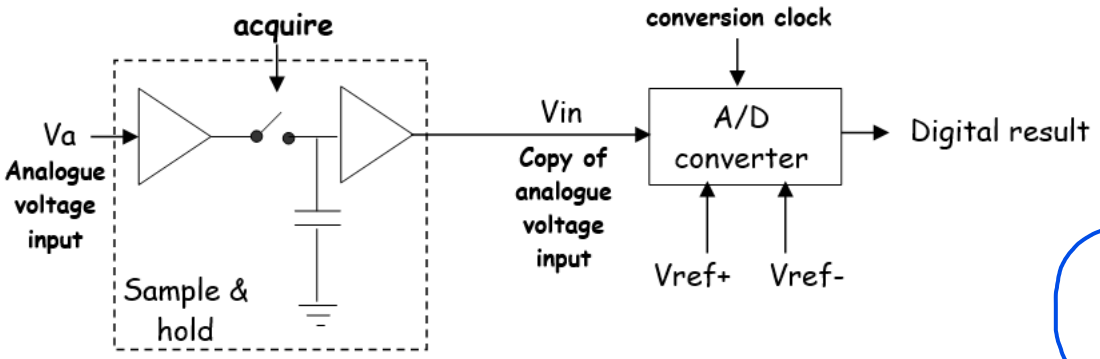


V(ADC) = 5 x(1000)/(1000+3000)
= 1.25

D

(a) 1.20 volts
(b) 5 volts
(c) 3.3 volts
(d) 1.25 volts

Q4.3



C

The purpose of a Sample/Hold circuit is to _____.

(a) to sample the analog signal, convert it to binary and hold it for the ADC
(b) to sample the analog signal, and hold it as a digital value for the ADC
(c) to sample the analog signal and hold it until the ADC finishes process the signal
(d) to sample the analog signal for the ADC and hold the converted digital value

Q4.4 A12-bit successive approximation ADC (SAR) operates with 3.3V and a 200 kHz clock. How long does the SAR ADC need to convert an input analogue signal of +2 volts?

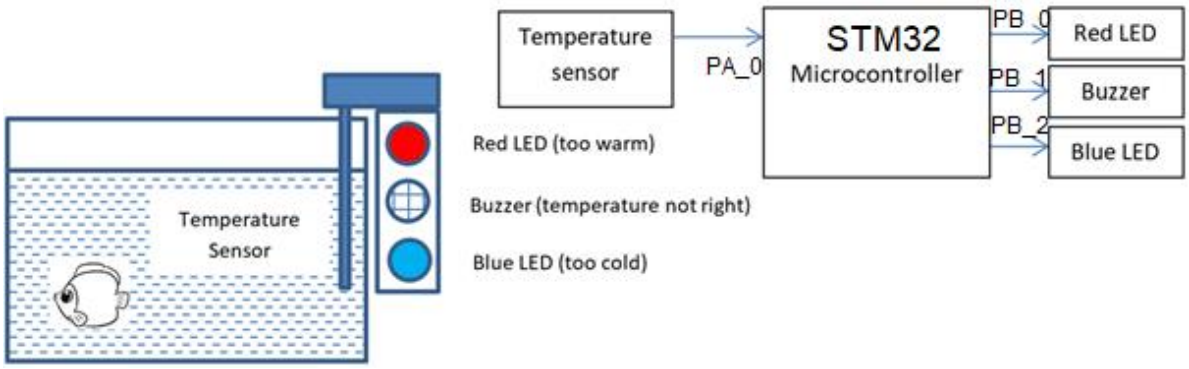
where $F = 200\,kHz$

$$T = \frac{1}{f}$$

$T = 5\,\mu s$

$12 \times 5\,\mu s = 60\,\mu s$

Q4.5. A water temperature sensor is used to monitor the water temperature in a fish tank, as follows:



The analog output of the temperature sensor is connected to the STM32 analog input channel PA_0.

The ADC result is right-justified and the converted digital value is then evaluated to control LEDs and Buzzer, according to the table below:

Temperature Sensor Output (Volts)	**Condition**	**Action**
< 1 volts	Too Cold	1. Turn ON buzzer 2. Turn ON Blue LED
≥ 1 volts and ≤ 2.5 volts	Normal	1. Turn OFF buzzer 2. Turn OFF Blue LED 3. Turn OFF Red LED
> 2.5 volts	Too Warm	1. Turn ON Buzzer 2. Turn ON Red LED

a) read() = Vin/Vref
= 1V/3.3V
= 0.3030
ADC values returned should be < 0.3030

read() = Vin/Vref
= 2.5V/3.3V
= 0.7576
ADC values returned should be > 0.7576

The C code running in the microcontroller is implemented in Table 1 below,

Code	Comments

#undef __ARM_FP
#include “mbed.h”
 	
AnalogIn temperature(PA_0);
DigitalOut Red(PB_0);
DigitalOut Blue(PB_2);
DigitalOut Buzzer(PB_1);
 	
int main( )
{ set_reference_voltage(3.
temperature.3);
 	
while(1)
{ temperature.read() > 0.7576
if(_____)
{ Red = 1;
Buzzer = 1;
}
temperature.read() < 0.3030
else if (_____)
{ Blue = 1;
Buzzer = 1;
}
 	
else Buzzer = 0;
{ Red = 0;
Blue = 0;
}
 	
}
 	
}
	// Line A = Set reference voltage
	
	// Line B = Check if temperature is too warm
	
	// Line C = Execute actions if water is too warm
	
	// Line D = Check if temperature is too cold
	
	// Line E = Execute actions if water is too cold
	
	// Water is normal temperature
	
	/* Line F = Execute actions if water is with normal temperature range */

Table 1

(a) Calculate the ADC values that returned by the AnalogIn class member function read(), when the water temperature sensor detects that the water is too cold and too warm.

(b) Complete the code in Table 1.